

Building a Minimum-Time Costate Library over the Phase Torus: Methods and Findings for the 70 mN DRO $\rightarrow$ Tulip Campaign

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Abstract

A costate library keyed by departure and arrival phase is a 24×24 grid of minimum-time transfers, each a certified root of the Pontryagin boundary-value problem. Filling every cell of the torus took eight rounds over three days and, more importantly, five distinct families of extremals that no single continuation could reach. This document explains the techniques that built it – the certification gate stack, pseudo-arclength continuation in the arrival phase, the departure-phase ribs, the family map, branch-blind direct solves for finding new families, cell-by-cell direct filling, and the second-order sweep – and what the family shares reported in the campaign log mean. The operational recipe is in `process/PHASE_TORUS_RUNBOOK.md`; the chronological record is FINDINGS §59–72.

1 The problem and the grid

The transfer is the planar-symmetric CR3BP Earth–Moon low-thrust minimum-time problem: a spacecraft of $m_0 = 150$ kg with a 70 mN, 900 s thruster leaves a distant retrograde orbit (period $\tau_D = 1.0$ nondimensional) and arrives on a 7-petal tulip orbit (branch $p = -1$, period $\tau_A = 2\pi(N_p - 2)/(N_p - 1) = 5.236$). Both orbits are periodic, so the boundary states are functions of a *phase* in $[0, 1)$: the departure state is $\mathbf{x}_D(s_D)$ and the arrival state is $\mathbf{x}_A(s_A)$. The pair (s_D, s_A) lives on a torus, and the library is the map $(s_D, s_A) \mapsto (t_f^*, \boldsymbol{\lambda}(0))$ on the lattice $s_D = k/24$, $s_A = 0.0754 + j/24$.

Each cell’s entry is a root $z = (\boldsymbol{\lambda}(0), t_f) \in \mathbb{R}^8$ of the shooting equations of the PMP boundary-value problem: seven costates at departure and the final time, such that the flown extremal (bang control $\mathbf{u} = -\boldsymbol{\lambda}_v/|\boldsymbol{\lambda}_v|$ at full throttle) lands on $\mathbf{x}_A(s_A)$ with the transversality condition $\lambda_m(t_f) = 0$. Roots come in *families* – connected branches in (s_D, s_A, z) space – and a given cell may lie under several of them; the library keeps the fastest *certified* one.

2 What “certified” means: the gate stack

Every entry in the library passed `certify_root`, which does more than converge the shooting equations:

1. multiple-shooting residual on $K = 24$ junctions below tolerance (`ms_tfmin`, homogeneous chart; the polish is capped, so a stalled Newton cannot hang the campaign);
2. the flown arrival, re-integrated from z alone with `ode113` at 10^{-13} , misses $\mathbf{x}_A(s_A)$ by less than a kilometre in position and by nothing measurable in velocity (“fly”);
3. transversality $\lambda_m(t_f) = 0$ read off that tight flight (an earlier gate read it off an `ode45` 10^{-10} flight and refused good roots – FINDINGS §59);

4. the free-time conjugate test (Bonnard–Caillaud–Trélat, the quotiented Jacobi determinant sampled through $(0, t_f)$): no interior sign change, near-misses resolved by a dense scan;
5. the sufficiency-hypothesis gates: strong Legendre $\min_t |\lambda_v| > 0$, $Q_{mt} > 0$, and normality with the abnormal-lift space of dimension one by a margin of at least $10\times$ (“lift margin”);
6. lunar clearance of 1900 km along the whole arc.

A root that fails any gate is recorded with its reason and never enters the catalog. This is what makes the availability grid honest: an empty cell means *no certified root found*, not *not tried*.

3 Pseudo-arclength continuation in the arrival phase

The arrival axis is where the sensitivity lives – the arrival state moves along a long, highly curved tulip petal – so it is walked by pseudo-arclength continuation (`arclength_ms`, `arclength_arrival`) rather than a fixed-step walker. The unknowns are the K junction states plus z , augmented by a normality scalar ρ (the homogeneous chart of `ms_tfmin_hom`), and the continuation parameter is s_A (“ q ”). At each step the tangent of the solution curve is computed from the Jacobian, a predictor step of length ds is taken along it, and a corrector Newton solve is done in the plane orthogonal to the tangent; ds adapts to the Newton count.

Three things the arc records are what the campaign lives on:

- **crossings** of the grid levels: whenever s_A passes a lattice value (possibly several times, unwrapped beyond $[0, 1)$), the root is re-solved at exactly that level and stored – these are the sheet’s candidates;
- **folds**: sign changes of the tangent’s s_A -component, localised and classified (rank-one loss of R_x with a regular augmented matrix) – where the family turns back in arrival phase;
- **the normality scalar** ρ along the walk: $\rho \rightarrow 0$ is the family losing normality (the abnormal direction opening), a different kind of end from a fold.

An arc of 4000 steps is 2–6 hours of solves. Since 2026-09-15 the arc so far is saved atomically every 50 steps (`.partialFile`), after a MATLAB segfault at step 2160 of one walk left nothing on disk; the partial is readable mid-walk and already answers “which columns will this family change”.

4 The arrival sheet

`sheet_from_arcs` assembles, per arrival phase, every candidate from every arc’s crossings and every seed (the library’s own certified roots, including the direct-found ones), certifies each, merges duplicates only when both t_f and z agree (keying on t_f alone once merged two distinct extremals), and takes the fastest certified candidate as the column’s *spine*. Nothing is discarded: a column keeps all its candidates with their verdicts, which is what lets the family map and the gap analysis work later.

5 Departure-phase ribs

Departure phase enters the problem only through the first junction’s fixed state, so the departure axis is cheap: `rib_from_crossing` holds s_A and walks s_D in steps of $1/24$ from the spine, re-solving from the previous point’s junctions and certifying every point; a failed step is bisected up to eight times. A rib stops – and says why – when a step cannot be certified even at $1/1728$ of a period: “normal-chart polish did not converge” (the root leaves the chart’s basin) or “dense conjugate scan not clear” (a conjugate point is approaching). Every cell beyond the stop is a hole for the direct filler.

Ribs run under a supervised work queue (`run_costate_library`, `campaign_supervisor.sh`): one unit per column, process-held kernel file locks, per-point checkpoints so a killed worker resumes mid-walk, heartbeats per solve stage, a parent supervisor per worker, and a finalizer that runs once when every column is published.

6 Families and the family map

A *family* is one connected branch of extremals. Operationally it is named by its *anchor* – the certified root the arcs start from – because the two arcs from an anchor (arrival phase increasing and decreasing) trace that branch. `family_map` reads, from the stored arcs, each family’s arrival-phase span, t_f span, folds, normality floor, and the *kind* of each end of its span: a fold, the walk budget (the branch continues, unmapped), a walk that ended with $\rho \rightarrow 0$, or the anchor itself (that direction never walked). It then attaches every certified root at every column to the arc that passes through it – t_f interpolated along the arc at that level, matched to 10^{-5} d; distinct roots at one phase sit ≥ 0.05 d apart – and every rib point to its spine root (the rib’s first three points extrapolated quadratically back to the spine). The catalog ships the map as `cat..families` and an `int8 family_index` per entry.

This is the meaning of the sentence in the campaign log, “*Five families now share the torus: fast 47%, direct18 22%, fast2 13%, direct11 13%, plus 24 cells on roots no arc has walked*” (round 6; the final numbers are in Table 1):

- the percentages are shares of the certified cells attributed to each family through the rib that produced them – 47% of the cells lie on ribs walked off spines of the original family;
- `family_index` = -1 , “roots no arc has walked”: certified spine roots at $s_D = 0$ that no arc passes through – the direct solver found them and no continuation has been anchored on them. Each is a candidate anchor; the `direct18` and `direct11` families were exactly such roots before their arcs were walked;
- `family_index` = -2 : rib points whose spine root the map could not identify – ribs from an earlier round whose spine is not among this sheet’s roots, and the direct-solved cells, which have no spine at all.

Table 1: The families of the finished torus (round 8). Spans are in arrival phase, unwrapped; “ t_f ” is the family’s range at $s_D = 0$.

family	anchor (s_A , t_f)	span	t_f (d)	ends	cells
fast	0.0754, 17.80 d	0.034–0.841	16.2–25.7	fold / fold	242 (42%)
A2	0.9087, 26.43 d	0.908–1.711	26.0–44.1	fold / budget	0
fast2	0.8671, 17.25 d	0.800–1.394	16.9–25.3	fold (S-bend) / fold	72 (12.5%)
direct18	0.7837, 17.83 d	0.573–1.224	16.8–38.2	fold / fold	121 (21%)
direct11	0.4921, 17.96 d	0.357–0.715	17.8–32.4	fold / stall	69 (12%)
roots no arc has walked (-1)					22 (3.8%)
rib points without an identified spine (-2)					50 (8.7%)

7 Finding families the continuation cannot reach

The central lesson of the campaign. Continuation follows its own sheet: from the original family’s anchor the arcs reached $s_A = 0.841$ and folded back, and the A2 family found at 0.9087 was 26 d – so

the first library reported 19 of 24 arrival phases and 22–26 d transfers on 0.6–0.9, all certified minima of *their* branches. The fast roots were found by *branch-blind direct solves*: a Hermite–Simpson collocation solve with Sundman regularisation (`casadi_mintime_dro`, $N = 800$, lunar clearance as a path constraint) at the phase in question, warm-started from the *other* family’s certified flight; the collocation solver does not know which branch it started on and lands in the basin of the fastest nearby extremal. Its solution is harvested into a multiple-shooting seed (`harvest_ms_seed`: covectors from the defect multipliers, sign vote) and certified by the same gate stack. That found 17–19 d roots across 0.78–1.03 (seven days under the sheet), then 0.4921 at 17.96 d (2.3 days under the sheet). Each became an anchor, its arcs were walked, the sheet was rebuilt with the new crossings, and the family map attributed every column.

A refused direct result is information too: at 0.5754 the fastest thing the solver finds rides the 1900 km clearance floor – a constrained arc, not an unconstrained extremal – so the 21.45 d root stands there.

8 Cell-by-cell direct filling and the improve pass

After the ribs, 38 cells were still empty. `fill_holes_direct` solves each hole directly, warm-started from up to three certified neighbours – the same column’s first, because they share the arrival geometry (a rib *is* this continuation), then the same row’s – harvests, certifies, keeps the fastest certified root and makes it a seed for the cells still to come, so a column chains like a rib. All 38 certified. Two lessons: a neighbouring-column seed at a different arrival phase sends the collocation solver to 100–400 d junk (cap the solve at 300 s; try the column first), and run under R2026a (the Parallel Computing Toolbox is not licensed under R2025b, and certification needs the pool).

The *improve pass* applies the same solve to filled cells more than 2 d slower than a column neighbour, seeded from faster neighbours only, keeping a root only if faster, and queueing the next slower neighbour when a cell improves. Column 15 ($s_A = 0.6587$), where the direct18 rib had stalled at once and the fast family’s 22–25 d rib filled the column, came down to 17.1–17.8 d over nine cells; nine cells above $s_D = 0.54$ resisted from both sides – a fold of the direct18 sheet in the departure phase, probably – and keep the 23 d family.

9 The second-order sweep and its sidecar

Every packaged entry is re-measured by `second_order_pass`: the conjugate determinant densely through $(0, t_f)$ (interior crossings, near-misses, multiplicity), the H6 margin $\lambda_m(0) < c/T$, the lift margin. The measurements are written back into the catalog and kept in a resumable *sidecar* keyed by cell and z ; when a catalog is re-packaged with more ribs the sidecar is merged – a record reused only for the entry whose cell and z match, everything else measured fresh – so a re-packaging costs only the changed entries.

10 The campaign loop and its monitoring

A round is: sheet from every arc and seed $\rightarrow$ ribs for the columns whose spine changed $\rightarrow$ package with every round’s ribs offered (the fastest certified point per cell wins) $\rightarrow$ audit (every entry re-flown and re-gated) $\rightarrow$ sweep. A new round starts whenever a new family changes spines (Section 7); a re-package without walking follows the direct filler. Eight rounds took the library from 406 entries and 19 columns to 576 of 576 (Table 2).

Long jobs write their product after every item (arcs: partial saves; ribs: per-point checkpoints; filler: per cell; sweep: per chunk), log to files (batch MATLAB buffers stdout), and are watched by scripts that exit on the job’s DONE line, its death, an error, a silent log or an hourly tick. A job at 0% CPU after its banner is diagnosed through its `matlab_helper` child – on 2026-09-15 a `git rev-parse` inside `system()` hung on the Xcode licence prompt for an hour; the chain now reads `.git` itself and trusts a process count only when it is made of integers.

Table 2: The rounds.

round	what changed	entries / verdict
1	first 24×24 from the original family	406, 19 columns
2	corrected transversality gate, supervised queue	409, audit 409/0
3	fast2 anchor and direct roots seeded	432
4	sheet from all six arcs and exact-phase seeds: 24/24 columns	515, audit 515/0
5	direct18 arcs: columns 14–17, 20 take 16.8–19.0 d	534, audit 534/0
6	direct11 arcs: columns 11–13 take 18.0–18.8 d	538, audit 538/0
7	+ <code>fill_holes_direct</code> , 38/38	576, audit 576/0
8	+ improve pass, merged sidecar	576/576, audit 576/0, sweep clean

11 Results

The finished library (Figure 1): 576 of 576 cells, t_f from 16.23 to 23.86 d (median 18.42 d); audit 576 ok / 0 bad; sweep 576 done with 0 interior conjugate crossings, worst H6 margin $4.44\times$, worst lift margin $10.6\times$. Every column is within 2.5 d of its own minimum except the nine stuck cells of column 15.

Open questions. (i) Whether the nine column-15 cells are true minima at 23 d or a fold the direct solver cannot jump – an arc in s_D at $s_A = 0.6587$ would say. (ii) The 22 spine roots no arc has walked (0.9921 at 18.30 d, 0.0337 at 18.14 d among them): arcs from them would attribute the last unmapped entries and might expose a sixth family. (iii) The fast sheet is not one branch: below the direct18 fold at 0.59 the direct11 family carries 0.49–0.60, and between them the cell at 0.5754 is clearance-limited; the topology of the fast sheet as a whole – how the five branches connect through their folds – is not yet drawn.

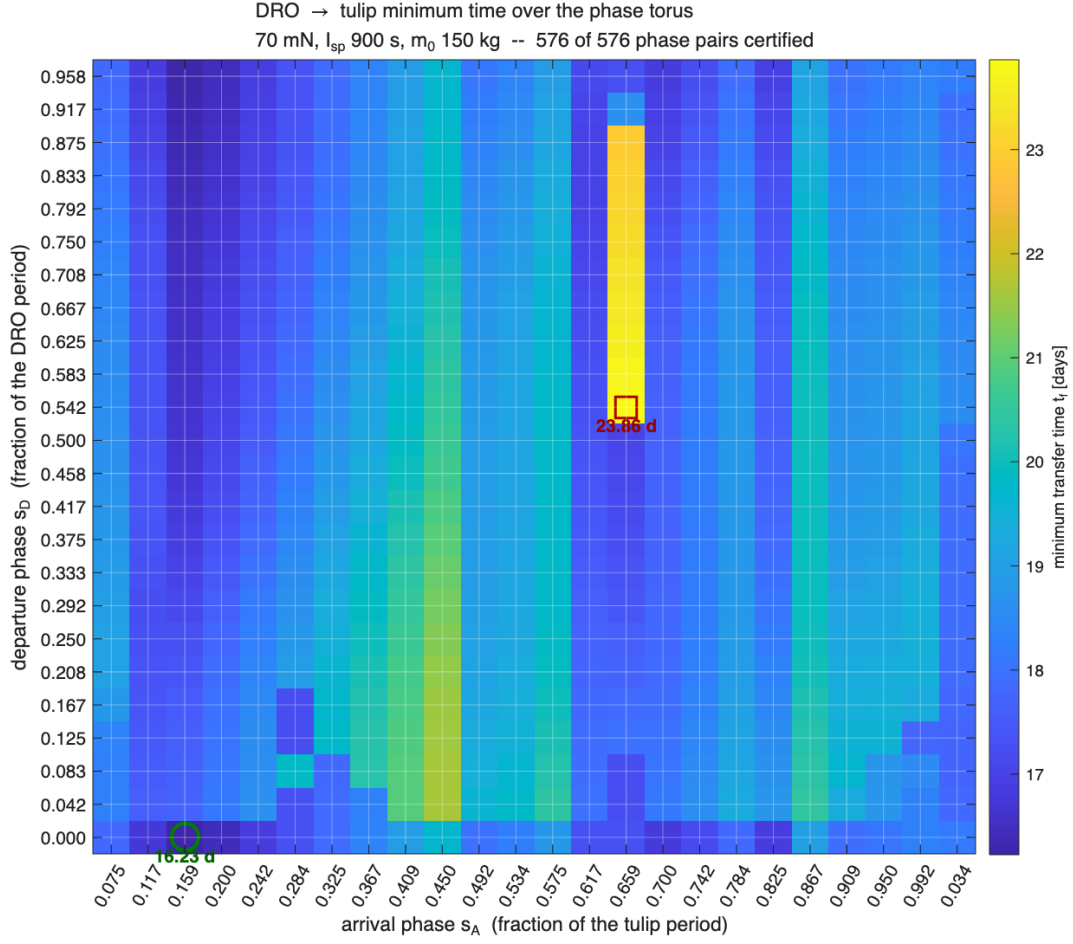


Figure 1: The phase torus: minimum time (days) over departure phase (rows) and arrival phase (columns), round 8. The ridge along $s_A \approx 0.45$ is the original family's slow end; the plateau above $s_A = 0.6$ is the direct18 and fast2 families; column 15's upper cells are the 23 d family the improve pass could not displace.